Station 1

1. Hematite (2)
2. Refer to answers below (0.5 each)
    a. False
    b. True
    c. True
    d. True
3. Fe_2O_3 (0.5)
4. Refer to answers below (1 each)
    a. Bauxite
    b. Azurite
    c. Aragonite
    d. Augite
    e. Aragonite
    f. Apatite
    g. Albite
    h. Barite
    i. Barite

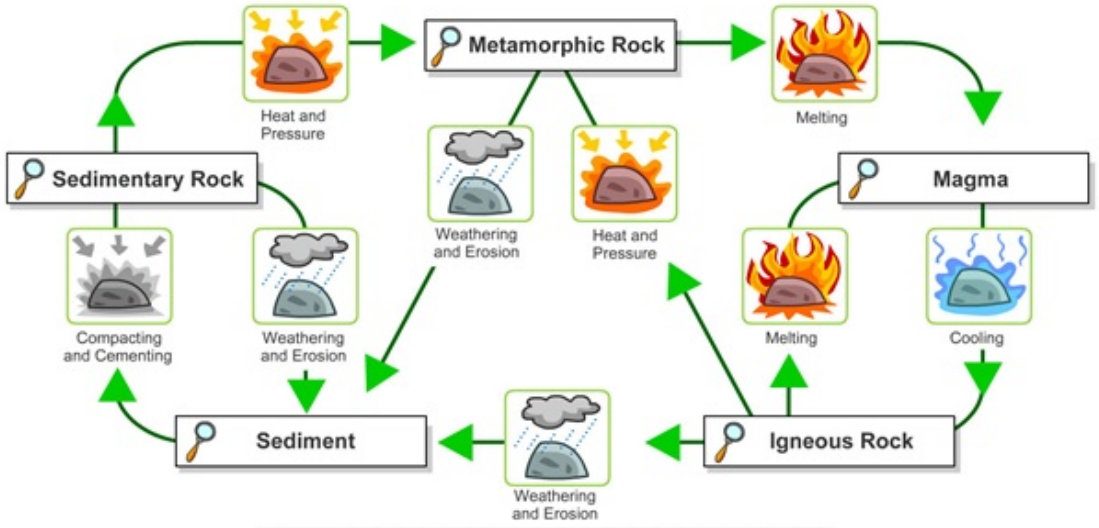
Station 2

1. The rock is mafic, cooled quickly, and had air bubbles. (3, 1 per underlined)
2. Areli went hiking and found a foliated rock with a fossil in it. The rock is
    a. sedimentary
    b. metamorphic
    c. igneous
    d. Volcanic
3. farther, closer (2, 1 per word)
4. grano-diorite (2)
5. iron (2)
6. a, d, f (3)

Station 3

1. A laccolith is a mushroom-shaped or lens-shaped igneous intrusion that forms when magma is injected between layers of sedimentary rock, causing the overlying strata to bulge upward. A batholith is a large, irregular mass of intrusive igneous rock that forces itself into surrounding strata. Batholiths have a surface area of at least 100 square kilometers (40 square miles), while laccoliths are typically smaller, with diameters usually less than 16 kilometers (10 miles).(4, 2 for shape, 2 for size)

2. Refer to chart below (5, 1 each)



3. less (2)
4. Joints are evenly spaced zones of weakness in massive igneous rock structures which form as the rock cools and contracts. (4)
5. Refer to the answers below. (5, 1 each)



Section 4

1. all the rocks were calcite (6, 1 each)
2. it would bubble (3)
3. Refer to answers below. (4, 1 each)
    a. white
    b. rhombohedron, perfect 3 directions
    c. 3
    d. 2.7
4. $CaCO_3$ (2)
5. a, d (2)
6. The blue color of blue calcite is usually due to the presence of various trace elements, especially copper.(2)

Section 5

1. Olivine (2)
2. carbon dioxide (3)
3. Peridot (3)
4. Refer to answers below (4.5, 1.5 each)
    a. a
    b. c
    c. b, c, e
5. Refer to the answers below (7, 1 each)
    1: laccolith
    2: dike
    3: batholith
    4: dike
    5: sill
    6: volcanic neck
    7: lopolith

Section 6

1. Family: Native Element, Diamond (2, 1 each)
2. Refer to the answers below
    a. False (1)
    b. Since Carbon-14 has a short half-life, most diamonds would be formed in a time range outside of the elements accessible time span. Instead, trace minerals found in the diamonds are used to date and analyze the diamond for geochronology, known as radiogenic isotope analysis. This is done by isolating other radioactive elements that have longer half-lives allowing scientists to determine the age of the diamond. (3)

3. A,d,e (2)

4. Diamonds are formed from carbon atoms being subjected to high pressure and heat environments, typically 140 to 190 km form the Earth's surface. Diamonds are also usually found in igneous rock, known as kimberlite deposits. These pipes, go though the crust to Earth's mantle and contain magma which inhibits diamonds. (3)

5. The four characteristics of diamonds that are used to determine the value of the diamond are carat weight, cut, color, and clarity. The GIA, Gemological Institue of America set these standard characteristics in 1953. (5, 1 for each characteristic identified, 1 for correct naming of institution)

6. Refer to the answers below (2, 1 each)
    a.) Triple point
    b.) Critical point

Section 7

1. a. igneous (1)
2. Refer to the answers below. (6, 0.5 each)

    A. obsidian white, b. Scoria white, c. syenite white, d. pegmatite white, e. Syenite white, f. tuff white

3. A mineral is an inorganic, natural occurring, solid, with a definite chemical composition, and a crystalline structure. A rock is a combination of minerals and potentially organic material, that can be formed though weathering, erosion, etc. (2)
4. Match the rocks shown above with the correct environment. Some words may be used more than once, multiple numbers for one answer, or not at all. Refer to the answers below. (6, 1 each)
   1. Batholiths, 2. Dikes, 3. Sills, 4. Laccoliths, 5. Volcanic necks, 6. Lava flows, 7. Pyroclastic flows
      A. 6, b. 7, c. 1,2,3,4, d. 1,2,3,4, e. 2,3,6,7, f. 7
5. c. aebfcd (2)

Section 8

1. d. Magnetite (2)
2. b. Magnetism (1)
3. Polymorphism in geology is regarded as the ability of a specimen to crystallize in multiple forms, even with the same chemical composition. It occurs due to the environment around a specimen constantly changing in pressure and temperature, causing different chemical and physical properties. (2)
4. Refer to the answers below. (5, 1 each)
   1. Contact metamorphism 2. Volcanic arc 3. Collisional mountain belt 4. Stable continent 5. Accretionary prism
5. a. Gneiss (2)
6. How does the grade of metamorphism affect the formation of a rock and give an example of each. The grade of metamorphism affects the formation of a rock because pressure and heat can vary leading to two different types of metamorphism, high grade and low grade. In low grade metamorphism, the original textures are preserved and typically have a fine grain size. In high grade, the rock undergoes a lot of heat and pressure leading to more physical changes in the texture and ten to have courser gain sizes. An example of high grade metamorphism is Schist and an example of low grade metamorphism is slate. (Answers may vary) (4)

Section 9

1. A,b,c,d (3)
2. Refer to the answers below. (3)
   The **Native Element** Family known for their uncombined chemical **elements** occur naturally and are typically more valuable for their industrial or jewelry uses.
3. The Silicate family is the largest group of minerals because of the abundance of silicon and oxygen in the Earth's crust, making up about 27.7% and 46% respectively. Additionally, due to the stong bond between these elements, this allows for a wide variety of minerals. (2)
4. Selenite (crystalline) (2)
5. Refer to the answers below (4, 1 each)
   a. Luster? vitreous/pearly
   b. Density? 2.317 g/cm^3

    c. Diaphaneity? transparent/translucent
    d. Chemical formula? $CaSO_4 \cdot 2H_2O$
6. Refer to the answers below. (2, 1 each)
    a. Discontinuous Series
    b. Continuous Series

Section 10

1. Refer to the answers below. (4, 1 each)
    a. Ultramafic
    b. Mafic
    c. Intermediate
    d. Felsic
2. c. Diorite (2)
3. d. Graphite (2)
4. a. Coquina Sedimentary rock, b. Silver Native Element Family, c. Calcite Carbonate Family, d. Talc Silicate family (8, 1 each)
5. Specimen c. Calcite is the one with the characteristic of double refraction. Double refraction occurs due to high birefringence (the diffferneces in the refractive index of the two rays of light) in calcite, which causes one ray of light to split into two. This then causes one object to look like two distant images, one each from the split ray of light. (3)